

PHY 607 Project1: ODE and Integral

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1 ODE: Heat exchange

1.1 Problem and numerical methods

For two objects with heat capacities C_1, C_2 and thermal coupling constant k , the temperatures satisfy

$$C_1 \frac{dT_1}{dt} = -k(T_1 - T_2), \quad C_2 \frac{dT_2}{dt} = k(T_1 - T_2). \quad (1.1)$$

The analytic solutions are

$$T_1(t) = \frac{C_1 T_1(0) + C_2 T_2(0)}{C_1 + C_2} + \frac{C_2 [T_1(0) - T_2(0)]}{C_1 + C_2} e^{-k(1/C_1 + 1/C_2)t}, \quad (1.2)$$

$$T_2(t) = \frac{C_1 T_1(0) + C_2 T_2(0)}{C_1 + C_2} - \frac{C_1 [T_1(0) - T_2(0)]}{C_1 + C_2} e^{-k(1/C_1 + 1/C_2)t}. \quad (1.3)$$

Both temperatures approach the same equilibrium temperature,

$$T_{\text{eq}} = \frac{C_1 T_1(0) + C_2 T_2(0)}{C_1 + C_2}, \quad (1.4)$$

and the total thermal energy $C_1 T_1 + C_2 T_2$ is conserved. I implemented the forward Euler method, the fourth-order Runge–Kutta method and SciPy’s `solve_ivp`.

1.2 Example results and validation

The parameters are set as

$$C_1 = 2, \quad C_2 = 3, \quad k = 0.5, \quad T_1(0) = 100, \quad T_2(0) = 20, \quad (1.5)$$

with $0 \leq t \leq 20$. Using a time step $h = 0.2$, RK4 gives

$$T_1(20) \simeq 52.01154, \quad T_2(20) \simeq 51.99231, \quad (1.6)$$

which agrees with the analytic solution to the displayed precision.

To compare the methods over the entire simulation, I calculated the maximum absolute error over all sampled times,

$$E_{\text{Euler}} = 0.762, \quad E_{\text{RK4}} = 7.61 \times 10^{-6}, \quad E_{\text{solve_ivp}} = 1.71 \times 10^{-2}. \quad (1.7)$$

where RK4 is much more accurate than forward Euler. The `solve_ivp` error is determined partly by the error tolerances supplied to SciPy.

Self-check: energy conservation First,

$$C_1 T_1(t) + C_2 T_2(t) \quad (1.8)$$

must remain constant. All three numerical solvers conserve this quantity to approximately 10^{-13} in the test case. See the project for more detail.

Self-check: equilibrium Second, the temperature difference should decrease toward zero. For RK4,

$$|T_1(0) - T_2(0)| = 80, \quad (1.9)$$

while at $t = 20$,

$$|T_1(20) - T_2(20)| \simeq 0.0192. \quad (1.10)$$

The numerical solution therefore approaches thermal equilibrium as expected.

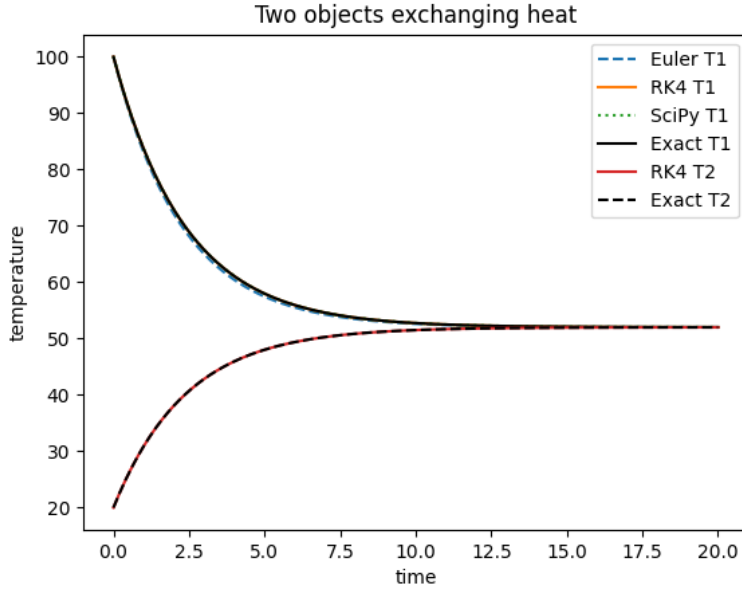


Figure 1. Temperature evolution for the parameters in Eq. (1.5) with $h = 0.2$. Euler, RK4, and SciPy are compared with the analytic solution for T_1 . RK4 and the analytic solution for T_2 are also shown.

Figure 1 shows the two temperatures approaching a common equilibrium. For these settings, Euler has a larger discrepancy from the analytic solution than RK4 or SciPy.

1.3 Truncation error and convergence

Euler has local truncation error $O(h^2)$ and global error $O(h)$. RK4 has local $O(h^5)$ and $O(h^4)$. Therefore, halving the step size should reduce the global error by approximately factors of 2 and 16, respectively.

Using $h = 0.125$ and $h = 0.0625$, I obtained

Method	Expected ratio $E(h)/E(h/2)$	Observed ratio
Euler	2	2.02
RK4	16	16.35

These agree with the expected orders.

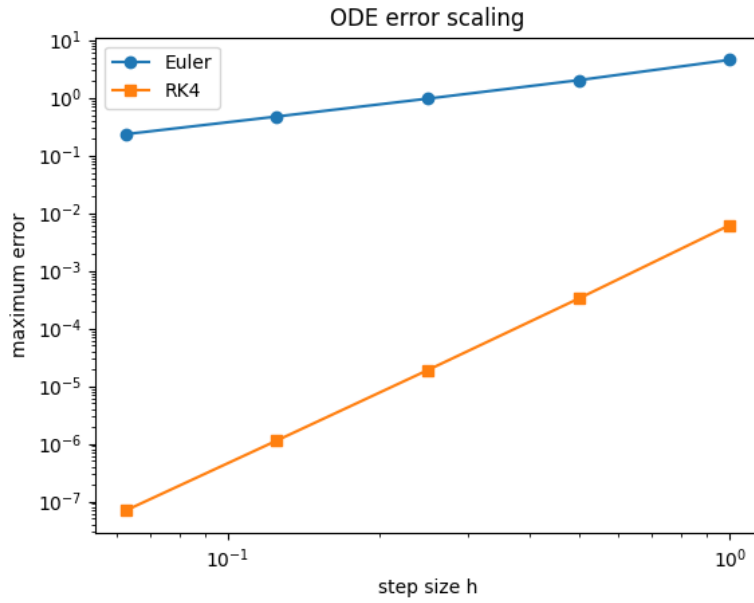


Figure 2. Maximum absolute temperature error over $0 \leq t \leq 20$ versus time step h .

Figure 2 shows that reducing the step size lowers the error. The convergence is consistent with global errors $O(h)$ for Euler and $O(h^4)$ for RK4.

2 Integral: Gaussian

2.1 Problem and numerical methods

The second problem is the Gaussian integral

$$I = \int_{\ell}^u e^{-a(x-\mu)^2} dx = \frac{\sqrt{\pi}}{2\sqrt{a}} [\operatorname{erf}(\sqrt{a}(u-\mu)) - \operatorname{erf}(\sqrt{a}(\ell-\mu))]. \quad (2.1)$$

I implemented the left Riemann sum, the trapezoid rule, and Simpson's rule. Simpson's rule requires an even number of intervals n . For an external comparison, I used SciPy's `quad`, which integrates the Gaussian using adaptive numerical quadrature.

2.2 Example results and validation

Set parameters as

$$a = 2, \quad \mu = 1, \quad \ell = -0.5, \quad u = 2.5, \quad n = 20. \quad (2.2)$$

The numerical results are

Method	Integral
Analytic	1.24993044
Left Riemann	1.24968274
Trapezoid	1.24968274
Simpson	1.24992156
SciPy quad	1.24993044

Self-check: translation symmetry First, shifting the integration interval and the Gaussian center by the same constant should not change the integral. In particular,

$$\int_{\ell}^u e^{-a(x-\mu)^2} dx = \int_{\ell+4}^{u+4} e^{-a[x-(\mu+4)]^2} dx. \quad (2.3)$$

The two numerical values differ by only

$$4.44 \times 10^{-16}. \quad (2.4)$$

Self-check: scaling symmetry Second, the substitution

$$v = \sqrt{a}(x - \mu) \quad (2.5)$$

gives

$$I = \frac{1}{\sqrt{a}} \int_{\sqrt{a}(\ell-\mu)}^{\sqrt{a}(u-\mu)} e^{-v^2} dv. \quad (2.6)$$

Evaluating the original and transformed integrals numerically gives a difference of only

$$2.22 \times 10^{-16}. \quad (2.7)$$

2.3 Truncation error and convergence

The left Riemann and trapezoid rules and Simpson's rule has global errors $O(h)$, $O(h^2)$, and $O(h^4)$ respectively.

I tested convergence using $n = 160$ and $n = 320$ for

$$\int_{-1}^2 e^{-x^2} dx. \quad (2.8)$$

Doubling n halves the step size, giving

Method	Expected ratio $E(h)/E(h/2)$	Observed ratio
Left Riemann	2	1.99
Trapezoid	4	4.00
Simpson	16	16.00

Figure 3 shows convergence consistent with global errors $O(h)$, $O(h^2)$, and $O(h^4)$ for the left Riemann, trapezoid, and Simpson methods, respectively.

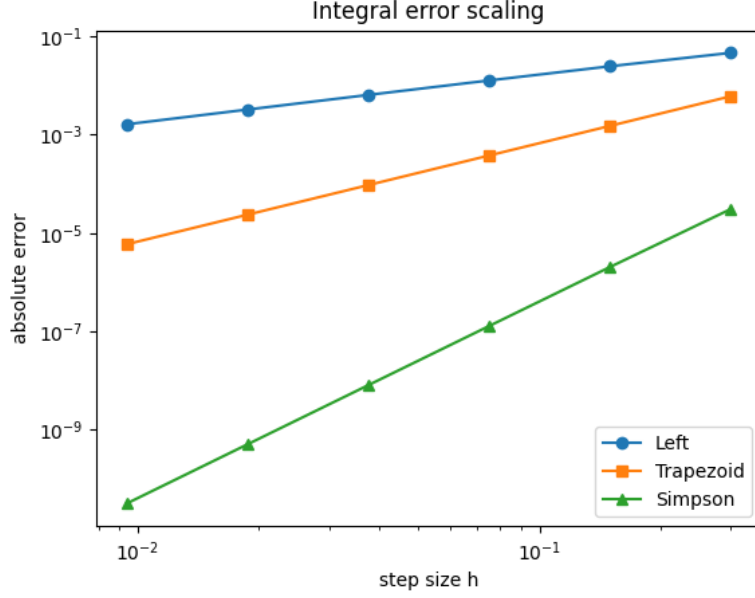


Figure 3. Absolute integration error relative to the analytic value of $\int_{-1}^2 e^{-x^2} dx$, plotted against $h = 3/n$ for $n = 10, 20, 40, 80, 160, 320$.

A Truncation errors

Here, local error means the error introduced in one time step or one integration panel. For ODE methods, the step starts from exact data.

Euler. Taylor expansion gives

$$y(t+h) = y(t) + hy'(t) + \frac{h^2}{2}y''(t) + O(h^3). \quad (\text{A.1})$$

Euler retains only the first two terms, so its local error is $O(h^2)$.

RK4. For the heat-exchange problem, the temperature difference $d = T_1 - T_2$ satisfies $d' = -\lambda d$, where $\lambda = k(1/C_1 + 1/C_2)$. Substituting this equation into the four RK4 stages gives

$$d_{i+1} = \left(1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}\right) d_i, \quad z = -\lambda h. \quad (\text{A.2})$$

The exact multiplier is e^z . Their difference begins with $z^5/120$, giving local error $O(h^5)$.

Left Riemann and trapezoid rules. Taylor expansion on one interval gives

$$\int_x^{x+h} f(s) ds - hf(x) = \frac{h^2}{2}f'(x) + O(h^3), \quad (\text{A.3})$$

$$\int_x^{x+h} f(s) ds - \frac{h}{2}[f(x) + f(x+h)] = -\frac{h^3}{12}f''(x) + O(h^4). \quad (\text{A.4})$$

Thus their local errors are $O(h^2)$ and $O(h^3)$, respectively.

Simpson's rule. For a panel of width $2h$ centered at m , Taylor expansion gives

$$\int_{m-h}^{m+h} f(x) dx = 2hf(m) + \frac{h^3}{3}f''(m) + \frac{h^5}{60}f^{(4)}(m) + O(h^7), \quad (\text{A.5})$$

$$\frac{h}{3}[f(m-h) + 4f(m) + f(m+h)] = 2hf(m) + \frac{h^3}{3}f''(m) + \frac{h^5}{36}f^{(4)}(m) + O(h^7). \quad (\text{A.6})$$

Subtracting gives $-h^5 f^{(4)}(m)/90 + O(h^7)$, so the local error is $O(h^5)$.

Global errors. Over a fixed interval there are $O(1/h)$ steps or panels. For stable ODE integration, propagation of the local errors gives global errors $O(h)$ for Euler and $O(h^4)$ for RK4. Summing the quadrature errors gives global errors $O(h)$, $O(h^2)$, and $O(h^4)$ for the left Riemann, trapezoid, and Simpson rules, respectively.